

PGR106

Linear Algebra

Lecture 5 – Matrix Equations $Ax = b$



Matrix-Vector Product

- A system of linear equations can be represented as a matrix-vector product in the form of $\mathbf{Ax} = \mathbf{b}$

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$
$$\begin{bmatrix} a_1 & a_2 & \cdot & \cdot & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} b \end{bmatrix}$$
$$\begin{array}{ccc} \underline{1 \times n} & \underline{n \times 1} & \underline{1 \times 1} \end{array}$$
$$\boxed{Ax = b}$$

Matrix-Vector Product

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\dots + \dots + \dots + \dots = \dots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

$$Ax = b$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{m1} & a_{m2} & \cdot & \cdot & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_m \end{bmatrix}$$

$$m \times n$$

$$n \times 1$$

$$m \times 1$$

Matrix-Vector Product

Example

$$x_1 - x_2 + 2x_3 = -4$$

$$x_2 - 3x_3 = 9$$

$$2x_1 + x_3 = 1$$

$$Ax = b$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -3 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 9 \\ 1 \end{bmatrix}$$

$$3 \times 3$$

$$3 \times 1$$

$$3 \times 1$$

Linear Combinations

- A system of linear equations can be represented as a linear combination of **variables** and **coefficients** from the columns of matrix **A** in the matrix-vector product of the form $\mathbf{Ax} = \mathbf{b}$

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ \dots &= \dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

$$Ax = b$$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{bmatrix}$$

$m \times n \qquad n \times 1 \qquad m \times 1$

Linear Combinations

- A system of linear equations can be represented as a linear combination of **variables** and **coefficients** from the columns of matrix **A** in the matrix-vector product of the form $\mathbf{Ax} = \mathbf{b}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{m1} & a_{m2} & \cdot & \cdot & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_m \end{bmatrix}$$

$m \times n$ $n \times 1$ $m \times 1$

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \cdot \\ \cdot \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \cdot \\ \cdot \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \cdot \\ \cdot \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_m \end{bmatrix}$$

Linear Combinations

Example

$$x_1 - x_2 + 2x_3 = -4$$

$$x_2 - 3x_3 = 9$$

$$2x_1 + x_3 = 1$$

$$Ax = b$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -3 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 9 \\ 1 \end{bmatrix}$$

$$3 \times 3$$

$$3 \times 1$$

$$3 \times 1$$

$$x_1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ 9 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ 0 \\ 2x_1 \end{bmatrix} + \begin{bmatrix} -x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} 2x_3 \\ -3x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 9 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 - x_2 + 2x_3 \\ x_2 - 3x_3 \\ 2x_1 + x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 9 \\ 1 \end{bmatrix}$$

Classification of Matrix-Vector Product

$$\begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{m1} & a_{m2} & \cdot & \cdot & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_m \end{bmatrix}$$

$m \times n \qquad n \times 1 \qquad m \times 1$

- **Case 1:**

Number of equations are equal to the number of variables

- **Case 2:**

Number of equations are less than the number of variables

- **Case 3:**

Number of equations are more than the number of variables

Classification of Matrix-Vector Product

- **Case 1:**

Number of equations are equal to the number of variables

$$\begin{aligned}5x_1 - x_2 + 2x_3 &= 8 \\ -2x_1 + 6x_2 + 9x_3 &= 15 \\ -7x_1 + 5x_2 - 3x_3 &= 7\end{aligned}$$

$$Ax = b$$

$$\begin{bmatrix} 5 & -1 & 2 \\ -2 & 6 & 9 \\ -7 & 5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 15 \\ 7 \end{bmatrix}$$

Classification of Matrix-Vector Product

- **Case 2:**

Number of equations are less than the number of variables

$$2x_1 + 3x_2 - 6x_3 + 2x_4 + 4x_5 = 5$$

$$4x_1 - 7x_2 + 8x_3 - 5x_4 + 2x_5 = 9$$

$$3x_1 - 9x_2 + 2x_3 - 4x_4 + 6x_5 = 7$$

$$Ax = b$$

$$\begin{bmatrix} 2 & 3 & -6 & 2 & 4 \\ 4 & -7 & 8 & -5 & 2 \\ 3 & -9 & 2 & -4 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \\ 7 \end{bmatrix}$$

Classification of Matrix-Vector Product

- **Case 3:**

Number of equations are more than the number of variables

$$2x_1 + 3x_2 = 5$$

$$4x_1 - 7x_2 = 9$$

$$3x_1 - 9x_2 = 7$$

$$6x_1 - 2x_2 = 8$$

$$Ax = b$$

$$\begin{bmatrix} 2 & 3 \\ 4 & -7 \\ 3 & -9 \\ 6 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \\ 7 \\ 8 \end{bmatrix}$$

Solving Systems of Linear Equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\dots + \dots + \dots + \dots = \dots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

$$Ax = b$$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$m \times n$$

$$n \times 1$$

$$m \times 1$$

Solving Systems of Linear Equations

- Using **Matrix Inverse**:

- Matrix A must be a **square** matrix.
- Determinant of A must **not** be **zero**.

$$Ax = b$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{m1} & a_{m2} & \cdot & \cdot & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_m \end{bmatrix}$$

$m \times n \qquad n \times 1 \qquad m \times 1$

$$Ax = b \quad \rightarrow \quad A^{-1}Ax = A^{-1}b \quad \rightarrow \quad Ix = A^{-1}b$$

$$x = A^{-1}b$$

Solving Systems of Linear Equations Using **Matrix Inverse**

Example

$$3x_1 + 5x_2 = 9$$

$$4x_1 + 7x_2 = 11$$

$$Ax = b$$

$$\begin{bmatrix} 3 & 5 \\ 4 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 11 \end{bmatrix}$$

$$2 \times 2$$

$$2 \times 1$$

$$2 \times 1$$

- ✓ A is a square matrix.
- ✓ $\det(A) = 3 * 7 - 5 * 4 = 1$

Solving Systems of Linear Equations Using **Matrix Inverse**

Example

$$3x_1 + 5x_2 = 9$$

$$4x_1 + 7x_2 = 11$$

$$Ax = b$$

$$\begin{bmatrix} 3 & 5 \\ 4 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 11 \end{bmatrix}$$

$2 \times 2 \qquad 2 \times 1 \qquad 2 \times 1$

$$x = A^{-1}b$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 4 & 7 \end{bmatrix}^{-1} \begin{bmatrix} 9 \\ 11 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{|A|} \begin{bmatrix} 7 & -5 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 9 \\ 11 \end{bmatrix} = \frac{1}{1} \begin{bmatrix} 7 & -5 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 9 \\ 11 \end{bmatrix} = \begin{bmatrix} 7 & -5 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 9 \\ 11 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 & -5 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 9 \\ 11 \end{bmatrix} = \begin{bmatrix} 7 * 9 + (-5) * 11 \\ -4 * 9 + 3 * 11 \end{bmatrix} = \begin{bmatrix} 63 - 55 \\ -36 + 33 \end{bmatrix} = \begin{bmatrix} 8 \\ -3 \end{bmatrix}$$

Solving Systems of Linear Equations Using **Matrix Inverse**

Exercise

Solve the following system of linear equations using matrix inverse.

$$2x + 3y = 8$$

$$4x + y = 10$$

Solving Systems of Linear Equations Using **Matrix Inverse**

Exercise

Solve the following system of linear equations using matrix inverse.

$$x + 2y + 3z = 9$$

$$2x + 3y + z = 8$$

$$3x + y + 2z = 7$$

Solving Systems of Linear Equations

- Using **Cramer's rule**:

- Matrix A must be a **square** matrix.
- Matrix A must be **invertible**.

$$Ax = b$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{m1} & a_{m2} & \cdot & \cdot & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ b_m \end{bmatrix}$$

$m \times n \qquad n \times 1 \qquad m \times 1$

$$A_i(b) = \begin{bmatrix} a_{11} & \cdot & b_1 & \cdot & a_{1n} \\ a_{21} & \cdot & b_2 & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{m1} & \cdot & b_n & \cdot & a_{mn} \end{bmatrix}$$


Column i

$$x_i = \frac{|A_i(b)|}{|A|}$$

$$\text{for } i = 1, 2, 3 \dots n$$

Solving Systems of Linear Equations Using **Cramer's Rule**

Example

$$3x_1 + 5x_2 = 9$$

$$4x_1 + 7x_2 = 11$$

$$A = \begin{bmatrix} 3 & 5 \\ 4 & 7 \end{bmatrix} \quad b = \begin{bmatrix} 9 \\ 11 \end{bmatrix}$$

$$|A| = 3 * 7 - 5 * 4 = 21 - 20 = 1$$

$$\underline{A_1(b)} = \begin{bmatrix} 9 & 5 \\ 11 & 7 \end{bmatrix}$$

$$|A_1(b)| = 9 * 7 - 5 * 11 = 63 - 55 = 8$$

$$x_1 = \frac{|A_1(b)|}{|A|} = \frac{8}{1} = \boxed{8}$$

$$\underline{A_2(b)} = \begin{bmatrix} 3 & 9 \\ 4 & 11 \end{bmatrix}$$

$$|A_2(b)| = 3 * 11 - 9 * 4 = 33 - 36 = -3$$

$$x_2 = \frac{|A_2(b)|}{|A|} = \frac{-3}{1} = \boxed{-3}$$

Solving Systems of Linear Equations Using **Cramer's Rule**

Exercise

Solve the following system of linear equations using Cramer's rule.

$$2x + 3y = 8$$

$$4x + y = 10$$

Solving Systems of Linear Equations Using **Cramer's Rule**

Exercise

Solve the following system of linear equations using Cramer's rule.

$$x + 2y + 3z = 9$$

$$2x + 3y + z = 8$$

$$3x + y + 2z = 7$$

End of Lecture 5



MATHEMATICS FOR MACHINE LEARNING

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